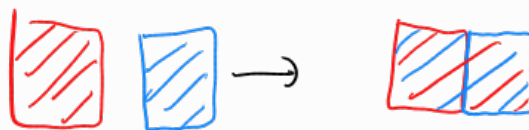


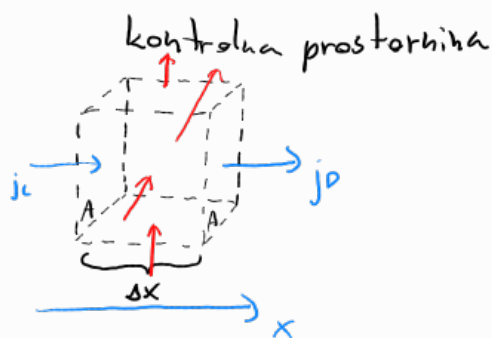
prenos snovi



prenos toplote

Difuzija snovi

poenostavitev: snovi imajo enako gostoto



$$\frac{\Delta m_i}{\Delta t} = -j_x^i A + j_x^i A = -(j_x^i - j_x^i) A$$

$$m_i = \rho_i A \Delta x$$

$$A \Delta x \frac{\partial \rho_i}{\partial t} = - \Delta j_x^i A$$

$$\frac{\partial \rho_i}{\partial t} = - \frac{\Delta j_x^i}{\Delta x} \Rightarrow \text{pospl.}$$

$$\rightarrow - \left(\frac{\Delta j_x}{\Delta x} + \frac{\Delta j_y}{\Delta y} + \frac{\Delta j_z}{\Delta z} \right) \rightarrow \Delta x, \Delta y, \Delta z \rightarrow 0$$

$$\rightarrow - \underbrace{\left(\frac{\partial j_x}{\partial x} + \frac{\partial j_y}{\partial y} + \frac{\partial j_z}{\partial z} \right)}_{= \nabla \cdot \vec{j}^i}$$

$\vec{j}^i = (j_x^i, j_y^i, j_z^i)$ vektor gostote snovnega toka snovi i

Kontinuitetna enačba:

$$\boxed{\frac{\partial \rho_i}{\partial t} = - \nabla \cdot \vec{j}^i}$$

$$\nabla \cdot \vec{j}^i = - \nabla \cdot (D \nabla \rho_i) = - D \nabla^2 \rho_i$$

če D ni odv. od r

$$D \left[\frac{\text{m}^2}{\text{s}} \right]$$

Fickov zakon: $\boxed{\vec{j}^i = -D \nabla \rho_i} =$

↑
difuzijska konst.

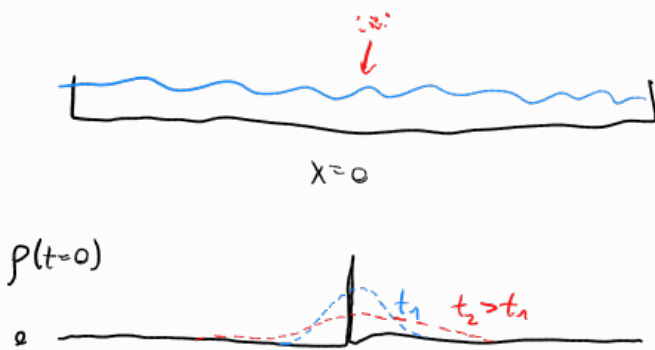
- Snovni tok $\vec{j}$, če $\nabla \rho_i \neq 0$
- lin. pribl. $\vec{j}^i \propto \nabla \rho_i$
- Snov teče tam, kjer je manj

Difuzijska enačba:

$$\frac{\partial \rho_i}{\partial t} = D \nabla^2 \rho_i$$

Lastnosti:

- linearna PDE
- parabolična PDE



rešitev: Greenova funkcija za dif. en.:

$$\rho(x,t) = \frac{C}{\sqrt{4\pi Dt}} \exp\left(-\frac{x^2}{4Dt}\right)$$

→ Gaussova funkc. za x , $\sigma_x = \sqrt{2Dt}$

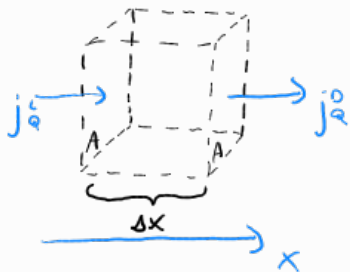
Časovni razvoj gostote:

• $\rho^i(x,0)$

$$\rho(x,t) = \int_{-\infty}^{\infty} \underbrace{\rho^i(x_0)}_{\text{zač. profil gostote}} \underbrace{\frac{1}{\sqrt{4\pi Dt}} \exp\left(-\frac{(x-x_0)^2}{4Dt}\right)}_{\text{Greenova funkc.}} dx_0$$

↖ integral po vsem prostoru

Difuzijska enačba za toploto



$$P = \frac{\partial Q}{\partial t} = \frac{\partial H}{\partial t} = A \Delta x \frac{\partial (\rho c_p T)}{\partial t} = A \Delta x \rho c_p \frac{\partial T}{\partial t}$$

$$-A(j_Q^{\text{in}} - j_Q^{\text{out}})$$

$$dH = m c_p dT$$

$$A \Delta x \rho$$

$$\left(\rho c_p \frac{\partial T}{\partial t} = - \frac{\Delta j_Q}{\Delta x} \right) \text{ kontinuitetna en.}$$

$$\Rightarrow \rho c_p \frac{\partial T}{\partial t} = - \nabla \cdot \vec{j}_Q \quad \vec{j}_Q = - \lambda \nabla T$$

↖ topl. prevodnost

$$\rightarrow \boxed{\frac{\partial T}{\partial t} = \frac{\lambda}{\rho c_p} \nabla^2 T} \text{ Difuzijska enačba}$$

$$D = \frac{\lambda}{\rho c_p} \text{ dif. konst. za toploto}$$

Križni transportni pojavi

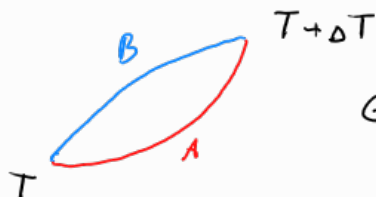
$$\nabla \rho_i \rightarrow \vec{j}_i$$

$$\nabla T \rightarrow \vec{j}_q$$

npr. $\nabla T \rightarrow$ shvni tok $\rightarrow$ el. tok

Termočlen

- 2 kovini A, B



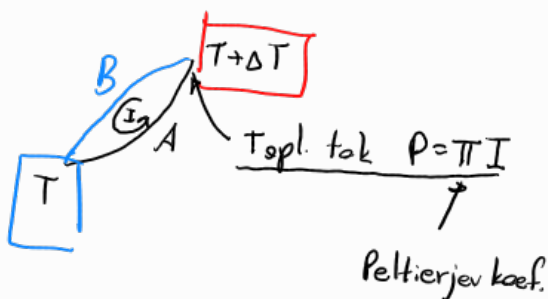
Gonilna napetost $U = a \Delta T$

↑
termo el. koef. ($1 \mu\text{V/K} \rightarrow 10 \text{mV/K}$)
kovine polprevodnik

Uporaba
- termometer

- če vmes damo breme - topl. stroj

Peltierjev pojav



Termočlen kot Carnotov topl. stroj

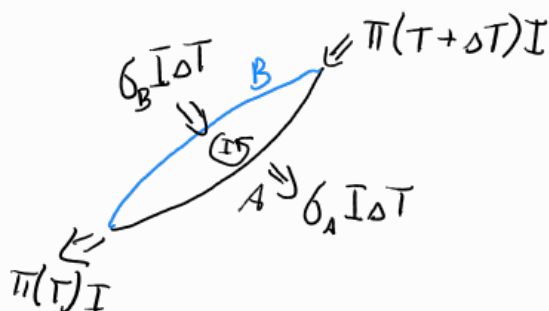
$$\eta = \frac{W}{Q} = \frac{UI t}{\pi I t} = \frac{U}{\pi}$$

$$\hookrightarrow \eta_{\text{carnot}} = 1 - \frac{T}{T + \Delta T} = \frac{T + \Delta T - T}{T + \Delta T} \approx \frac{\Delta T}{T}$$

$\Delta T \ll T$

$$\frac{U}{\pi} = \frac{a \Delta T}{\pi} \approx \frac{\Delta T}{T} \rightarrow \frac{a}{\pi} = \frac{1}{T}, \quad \boxed{\pi = aT}$$

zveza med
 π in a



Topl. bilanca?

$$\pi(T + \Delta T)I - \pi(T)I + \sigma_B I \Delta T - \sigma_A I \Delta T - (R_A + R_B)I^2 = 0$$

$$\hookrightarrow \pi(T) + \frac{d\pi}{dT} \Delta T$$

Jouleova
toplota

$$\frac{d\pi}{dT} \Delta T + (\sigma_B - \sigma_A) \Delta T - \underbrace{(R_A + R_B)I^2}_{U = a \Delta T} = 0 \quad / : \Delta T$$

$$\boxed{\frac{d\pi}{dT} + \sigma_B - \sigma_A - a = 0} \quad \text{spl. zveza med } \pi \text{ in } a.$$